

# Secure Secrets, the CLI Way

Student Lab: Azure Key Vault, Blob Storage & a .NET API

*All commands use Windows Command Prompt (cmd.exe)*

## Lab Overview

In this lab you will build a complete, security-first cloud application entirely from the command line. By the end you will have an Azure Key Vault storing a Blob Storage connection string, and an ASP.NET Core Web API that retrieves that secret at startup using a Managed Identity — no passwords, no hard-coded credentials, and no Portal clicking.

### What You Will Build

| Lab                                 | Goal                                                                                 |
|-------------------------------------|--------------------------------------------------------------------------------------|
| Lab 1: CLI Setup                    | Sign in, set variables, create the resource group                                    |
| Lab 2: Key Vault                    | Create an RBAC-enabled vault and test secret read/write                              |
| Lab 3: Blob Storage                 | Create a storage account and store the connection string in Key Vault                |
| Lab 4: .NET API Scaffold            | Scaffold an ASP.NET Core Web API and add the Azure packages                          |
| Lab 5: Key Vault Integration        | Wire Key Vault into the app configuration using DefaultAzureCredential               |
| Lab 6: Blob Endpoints               | Build upload and download endpoints that use the Key-Vault-sourced connection string |
| Lab 7: Deploy with Managed Identity | Deploy the API to App Service and grant it passwordless access to Key Vault          |
| Lab 8: Cleanup                      | Delete all resources with a single command                                           |

### Before You Start

- Open a Windows Command Prompt (cmd.exe). Look for plain C:\...> at the prompt — if you see PS C:\...> you are in PowerShell and the variable syntax will be different.
- Confirm the Azure CLI is installed: `az --version`
- Confirm the .NET SDK is installed: `dotnet --version` (8.0 or later required)
- You need an active Azure subscription.

#### Tip

All variables you set with the `set` command only last for the current session. Keep this `cmd.exe` window open for the whole lab. If you close it and reopen, you will need to re-run the `set` commands from Lab 1.

#### Important

Key Vault names and Storage account names must be globally unique across all of Azure. Use the `%RANDOM%` trick in Lab 1 to generate a unique suffix — do not change it once you have set it.

## Lab 1: CLI Setup

### Step 1 — Sign In

1. Run the following command. A browser window will open for you to log in.

```
az login
```

```
vaibhavgupta@FVFG905WQ05F-vaibhavgupta Day 48 July 1 % az login
A web browser has been opened at https://login.microsoftonline.com/organizations/oauth2/v2.0/authorize. Please continue the login in the web browser. If no web browser is available or if the web browser fails to open, use device code flow with 'az login --use-device-code'.

Retrieving tenants and subscriptions for the selection...
The following tenants don't contain accessible subscriptions. Use 'az login --allow-no-subscriptions' to have tenant level access.
7f7268a4-7179-4a42-9837-40255b1c21e5 'Presidio Network Solutions'

[Tenant and subscription selection]

No      Subscription name      Subscription ID      Tenant
-----
[1] *   Training-2026      e25db867-f63d-489a-8bce-7de21e5f94a1  Presidio Networked Solutions

The default is marked with an *; the default tenant is 'Presidio Networked Solutions' and subscription is 'Training-2026' (e25db867-f63d-489a-8bce-7de21e5f94a1).

Select a subscription and tenant (Type a number or Enter for no changes):

Tenant: Presidio Networked Solutions
Subscription: Training-2026 (e25db867-f63d-489a-8bce-7de21e5f94a1)

[Announcements]
With the new Azure CLI login experience, you can select the subscription you want to use more easily. Learn more about it and its configuration at https://go.microsoft.com/fwlink/?linkid=2271236

If you encounter any problem, please open an issue at https://aka.ms/azclibug

[Warning] The login output has been updated. Please be aware that it no longer displays the full list of available subscriptions by default.

vaibhavgupta@FVFG905WQ05F-vaibhavgupta Day 48 July 1 %
```

### Step 2 — Set Your Variables

2. Type each command exactly as shown. Do not add spaces around the = sign.

```
REM Set reusable variables (cmd.exe)
set RG=cli-lab-rg
set LOCATION=eastus
set /A SUFFIX=%RANDOM%
set KV_NAME=cli-lab-kv-%SUFFIX%
set SA_NAME=clilabsa%SUFFIX%
set CONTAINER_NAME=uploads
```

3. Confirm each variable was set correctly:

```
echo %RG%
echo %KV_NAME%
echo %SA_NAME%
```

```
vaibhavgupta@FVF9985WQ85F-vaibhavgupta Day 48 July 1 % export RG="cli-lab-rg"
export LOCATION="eastus"
export SUFFIX=$RANDOM
export KV_NAME="cli-lab-kv-$SUFFIX"
export SA_NAME="clilabsa$SUFFIX"
export CONTAINER_NAME="uploads"
vaibhavgupta@FVF9985WQ85F-vaibhavgupta Day 48 July 1 % echo $RG
echo $KV_NAME
echo $SA_NAME
cli-lab-rg
cli-lab-kv-3832
clilabsa3832
vaibhavgupta@FVF9985WQ85F-vaibhavgupta Day 48 July 1 % █
```

Write your values here for reference during the rest of the lab:

**KV\_NAME value**

cli-lab-kv-3832

**SA\_NAME value**

clilabsa3832

**SUFFIX value**

### Step 3 — Create the Resource Group

```
az group create --name %RG% --location %LOCATION%
```

```
clilabsa3832
vaibhavgupta@FVF9985WQ85F-vaibhavgupta Day 48 July 1 % az group create --name VaibhavResourceGroup --location $LOCATION
/opt/homebrew/Cellar/azure-cli/2.87.0/libexec/lib/python3.13/site-packages/azure/mgmt/resource/deploymentstacks/models/_models.py:119: SyntaxWarning: invalid escape sequence '\/'
following actions are automatically appended to 'excludedActions': '\read' and
{
  "id": "/subscriptions/e25db867-f63d-489a-8bce-7de21e5f94a1/resourceGroups/VaibhavResourceGroup",
  "location": "eastus",
  "managedBy": null,
  "name": "VaibhavResourceGroup",
  "properties": {
    "provisioningState": "Succeeded"
  },
  "tags": null,
  "type": "Microsoft.Resources/resourceGroups"
}
vaibhavgupta@FVF9985WQ85F-vaibhavgupta Day 48 July 1 % █
```

## Lab 2: Create and Secure a Key Vault

### Step 1 — Create the Vault

4. Run the following command. The ^ symbol continues the command on the next line — type it exactly as shown.

```
az keyvault create ^
--name %KV_NAME% ^
--resource-group %RG% ^
--location %LOCATION% ^
--enable-rbac-authorization true
```

```
vaibhavgupta@FVFG905WQ05F-vaibhavgupta Day 48 July 1 % az keyvault create \
--name $KV_NAME \
--resource-group $RG \
--location $LOCATION \
{
  --enable-rbac-authorization true
}
{
  "id": "/subscriptions/e25db867-f63d-489a-8bce-7de21e5f94a1/resourceGroups/cli-lab-rg/providers/Microsoft.KeyVault/vaults/cli-lab-kv-3832",
  "location": "eastus",
  "name": "cli-lab-kv-3832",
  "properties": {
    "accessPolicies": [],
    "createMode": null,
    "enablePurgeProtection": null,
    "enableRbacAuthorization": true,
    "enableSoftDelete": true,
    "enabledForDeployment": false,
    "enabledForDiskEncryption": false,
    "enabledForTemplateDeployment": false,
    "hsmPoolResourceId": null,
    "networkAcls": null,
    "privateEndpointConnections": null,
    "provisioningState": "Succeeded",
    "publicNetworkAccess": "Enabled",
    "sku": {
      "family": "A",
      "name": "standard"
    },
    "softDeleteRetentionInDays": 90,
    "tenantId": "2a731c61-a2b2-4661-8409-5b861cf40d0c",
    "vaultUri": "https://cli-lab-kv-3832.vault.azure.net/"
  },
  "resourceGroup": "cli-lab-rg",
  "systemData": {
    "createdAt": "2026-07-01T09:43:36.819000+00:00",
    "createdBy": "vgupta1@presidio.com",
    "createdByType": "User",
    "lastModifiedAt": "2026-07-01T09:43:36.819000+00:00",
    "lastModifiedBy": "vgupta1@presidio.com",
    "lastModifiedByType": "User"
  },
  "tags": {},
  "type": "Microsoft.KeyVault/vaults"
}
vaibhavgupta@FVFG905WQ05F-vaibhavgupta Day 48 July 1 %
vaibhavgupta@FVFG905WQ05F-vaibhavgupta Day 48 July 1 %
vaibhavgupta@FVFG905WQ05F-vaibhavgupta Day 48 July 1 % █
```

### Step 2 — Grant Yourself Access

5. Capture your user principal name and subscription ID, then create the role assignment:

```
for /f "delims=" %a in ('az ad signed-in-user show --query userPrincipalName -o tsv') do set MY_UPN=%a
for /f "delims=" %a in ('az account show --query id -o tsv') do set SUBSCRIPTION_ID=%a

az role assignment create ^
  --role "Key Vault Secrets Officer" ^
  --assignee %MY_UPN% ^
  --scope
"/subscriptions/%SUBSCRIPTION_ID%/resourceGroups/%RG%/providers/Microsoft.KeyVault/vaults/%KV_NAME%"
```

**Tip**

The for /f loop captures command output into a variable. Use single % signs when typing at the prompt (not inside a .bat file).

**Important**

Role assignments can take 1–2 minutes to propagate. If the next step returns a 403 error, wait a minute and try again.

```
zsh: parse error near `/'
vaibhavgupta@FVF0905WQ05F-vaibhavgupta Day 48 July 1 % MY_UPN=$(az ad signed-in-user show --query userPrincipalName -o tsv)
SUBSCRIPTION_ID=$(az account show --query id -o tsv)

az role assignment create \
  --role "Key Vault Secrets Officer" \
  --assignee "$MY_UPN" \
  --scope "/subscriptions/$SUBSCRIPTION_ID/resourceGroups/$RG/providers/Microsoft.KeyVault/vaults/$KV_NAME"
[AuthorizationFailed] The client 'vgupta1@presidio.com' with object id '18f12757-ddc1-413f-88db-4ed9843b2389' does not have authorization to perform action 'Microsoft.Authorization/roleAssignments/write' over scope '/subscriptions/e25db867-f63d-489a-8bce-7de21e5f94a1/resourceGroups/cli-lab-rg/providers/Microsoft.KeyVault/vaults/cli-lab-kv-3832/providers/Microsoft.Authorization/roleAssignments/7fed057c-6490-48c4-b2f4-ad53bc34cf32' or the scope is invalid. If access was recently granted, please refresh your credentials.
Code: AuthorizationFailed
Message: The client 'vgupta1@presidio.com' with object id '18f12757-ddc1-413f-88db-4ed9843b2389' does not have authorization to perform action 'Microsoft.Authorization/roleAssignments/write' over scope '/subscriptions/e25db867-f63d-489a-8bce-7de21e5f94a1/resourceGroups/cli-lab-rg/providers/Microsoft.KeyVault/vaults/cli-lab-kv-3832/providers/Microsoft.Authorization/roleAssignments/7fed057c-6490-48c4-b2f4-ad53bc34cf32' or the scope is invalid. If access was recently granted, please refresh your credentials.
vaibhavgupta@FVF0905WQ05F-vaibhavgupta Day 48 July 1 %
```

## Step 3 — Smoke-Test the Vault

- Set a test secret and read it back:

```
az keyvault secret set --vault-name %KV_NAME% --name "HelloWorld" --value "it-works"
az keyvault secret show --vault-name %KV_NAME% --name "HelloWorld" --query value -o tsv
```

The second command should print: it-works

- Delete the test secret:

```
az keyvault secret delete --vault-name %KV_NAME% --name "HelloWorld"
```

# RBAC issue, no permissions

```
vaibhavgupta@FVF098WQ65F--vaibhavgupta Day 48 July 1 % az keyvault secret delete \
  --vault-name "SKV_NAME" \
  --name "HelloWorld"
Warning! If you have soft-delete protection enabled on this key vault, this secret will be moved to the soft deleted state. You will not be able to create a secret with the same name within this key vault until the secret has been purged from the soft-deleted state. Please see the following documentation for additional guidance.
https://learn.microsoft.com/azure/key-vault/general/soft-delete-overview
(Forbidden) Caller is not authorized to perform action on resource.
If role assignments, deny assignments or role definitions were changed recently, please observe propagation time.
Caller: appid=04b87795-8ddb-461a-bbee-02f9e1b77b46;oid=18f12757-ddc1-413f-88db-4ed9843b2309;iss=https://sts.windows.net/2a731c61-a2b2-4661-8409-5b861cf40d8c/
Action: 'Microsoft.KeyVault/vaults/secrets/delete'
Resource: '/subscriptions/e25db867-f63d-489a-8bce-7de21e5f94a1/resourcegroups/cli-lab-rg/providers/microsoft.keyvault/vaults/cli-lab-kv-3832/secrets/helloworld'
Assignment: (not found)
DenyAssignmentId: null
DecisionReason: null
Vault: cli-lab-kv-3832;location=eastus

Code: Forbidden
Message: Caller is not authorized to perform action on resource.
If role assignments, deny assignments or role definitions were changed recently, please observe propagation time.
Caller: appid=04b87795-8ddb-461a-bbee-02f9e1b77b46;oid=18f12757-ddc1-413f-88db-4ed9843b2309;iss=https://sts.windows.net/2a731c61-a2b2-4661-8409-5b861cf40d8c/
Action: 'Microsoft.KeyVault/vaults/secrets/delete'
Resource: '/subscriptions/e25db867-f63d-489a-8bce-7de21e5f94a1/resourcegroups/cli-lab-rg/providers/microsoft.keyvault/vaults/cli-lab-kv-3832/secrets/helloworld'
Assignment: (not found)
DenyAssignmentId: null
DecisionReason: null
Vault: cli-lab-kv-3832;location=eastus

Inner error: {
  "code": "ForbiddenByRbac"
}
vaibhavgupta@FVF098WQ65F--vaibhavgupta Day 48 July 1 %
```

## Lab 3: Blob Storage and the Connection String Secret

### Step 1 — Create the Storage Account and Container

```
az storage account create ^
--name %SA_NAME% ^
--resource-group %RG% ^
--location %LOCATION% ^
--sku Standard_LRS ^
--kind StorageV2

az storage container create ^
--name %CONTAINER_NAME% ^
--account-name %SA_NAME% ^
--auth-mode login
```

```
"isSftpEnabled": null,
"isSkuConversionBlocked": null,
"keyCreationTime": {
  "key1": "2026-07-01T09:55:09.429677+00:00",
  "key2": "2026-07-01T09:55:09.429677+00:00"
},
"keyPolicy": null,
"kind": "StorageV2",
"largeFileSharesState": null,
"lastGeoFailoverTime": null,
"location": "eastus",
"minimumTlsVersion": "TLS1_0",
"name": "c1ilabsa3832",
"networkRuleSet": {
  "bypass": "None",
  "defaultAction": "Allow",
  "ipRules": [],
  "ipv6Rules": [],
  "resourceAccessRules": null,
  "virtualNetworkRules": []
},
"placement": null,
"primaryEndpoints": {
  "blob": "https://c1ilabsa3832.blob.core.windows.net/",
  "dfs": "https://c1ilabsa3832.dfs.core.windows.net/",
  "file": "https://c1ilabsa3832.file.core.windows.net/",
  "internetEndpoints": null,
  "ipv6Endpoints": null,
  "microsoftEndpoints": null,
  "queue": "https://c1ilabsa3832.queue.core.windows.net/",
  "table": "https://c1ilabsa3832.table.core.windows.net/",
  "web": "https://c1ilabsa3832.z13.web.core.windows.net/"
},
"primaryLocation": "eastus",
"privateEndpointConnections": [],
"provisioningState": "Succeeded",
"publicNetworkAccess": null,
"resourceGroup": "c1l-lab-rg",
"routingPreference": null,
"sasPolicy": null,
"secondaryEndpoints": null,
"secondaryLocation": null,
"sku": {
  "name": "Standard_LRS",
  "tier": "Standard"
},
"statusOfPrimary": "available",
"statusOfSecondary": null,
"storageAccountSkuConversionStatus": null,
"systemData": null,
"tags": {},
"type": "Microsoft.Storage/storageAccounts",
"zones": null
}
{
  "created": true
}
vaibhavgupta@FVG985WQ85F--vaibhavgupta Day 48 July 1 % []
```

### Step 2 — Capture the Connection String

```
for /f "delims=" %a in ('az storage account show-connection-string --name %SA_NAME% --
resource-group %RG% --query connectionString -o tsv') do set CONN_STRING=%a
```



**Tip**

Verify the variable was captured: `echo %CONN_STRING%` — it should start with `DefaultEndpointsProtocol=https`

**Step 3 — Store the Connection String in Key Vault**

```
az keyvault secret set ^
--vault-name %KV_NAME% ^
--name "BlobStorageConnectionString" ^
--value "%CONN_STRING%"
```

```
vaibhavgupta@FVF0995WQ65F-vaibhavgupta Day 48 July 1 % echo %CONN_STRING%
DefaultEndpointsProtocol=https;EndpointSuffix=core.windows.net;AccountName=clilabsa3832;AccountKey=zcSqhn/F0Irc80Gp6k6/KE+9dGMwTIK8RW0vWl0LnVYf1Nk0mpgt+tfrNlwj6beb8qIp6dN7n+AST2+IDbQ==;BlobEndpoint=https://clilabsa3832.blob.core.windows.net;/FileEndpoint=https://clilabsa3832.file.core.windows.net;/QueueEndpoint=https://clilabsa3832.queue.core.windows.net;/TableEndpoint=https://clilabsa3832.table.core.windows.net/
vaibhavgupta@FVF0995WQ65F-vaibhavgupta Day 48 July 1 % az keyvault secret set \
--vault-name "SKV_NAME" \
--name "BlobStorageConnectionString" \
--value "%CONN_STRING%"
[
  {
    "code": "Forbidden",
    "message": "Caller is not authorized to perform action on resource.",
    "details": {
      "Caller": "appid=84b87795-8ddb-461a-bbee-02f9e1bf7b44;oid=18f12757-ddc1-413f-88db-4ed9843b2389;iss=https://sts.windows.net/2a731c61-a2b2-4661-8489-5b861cf40d8c/",
      "Action": "Microsoft.KeyVault/vaults/secrets/setSecret/action",
      "Resource": "/subscriptions/e25db867-f63d-489a-8bce-7de21e5f94a1/resourcegroups/cli-lab-rg/providers/microsoft.keyvault/vaults/cli-lab-kv-3832/secrets/blobstorageconnectionstring",
      "Assignment": "(not found)",
      "DenyAssignmentId": null,
      "DecisionReason": null,
      "Vault": "cli-lab-kv-3832;location=eastus"
    }
  },
  {
    "code": "Forbidden",
    "message": "Caller is not authorized to perform action on resource.",
    "details": {
      "Caller": "appid=84b87795-8ddb-461a-bbee-02f9e1bf7b44;oid=18f12757-ddc1-413f-88db-4ed9843b2389;iss=https://sts.windows.net/2a731c61-a2b2-4661-8489-5b861cf40d8c/",
      "Action": "Microsoft.KeyVault/vaults/secrets/setSecret/action",
      "Resource": "/subscriptions/e25db867-f63d-489a-8bce-7de21e5f94a1/resourcegroups/cli-lab-rg/providers/microsoft.keyvault/vaults/cli-lab-kv-3832/secrets/blobstorageconnectionstring",
      "Assignment": "(not found)",
      "DenyAssignmentId": null,
      "DecisionReason": null,
      "Vault": "cli-lab-kv-3832;location=eastus"
    }
  }
]
Inner error: {
  "code": "ForbiddenByRBAC"
}
vaibhavgupta@FVF0995WQ65F-vaibhavgupta Day 48 July 1 %
```

Permission issue

**Important**

This is the last time the connection string will be handled directly. From here on, only Key Vault knows it.

## Lab 4: Scaffold the .NET Web API

### Step 1 — Create the Project

#### Tip

Choose a folder on your machine where you want to create the project, then cd into it before running these commands.

```
dotnet new webapi -n BlobVaultApi --use-controllers
cd BlobVaultApi
```

### Step 2 — Add the Azure NuGet Packages

```
dotnet add package Azure.Identity
dotnet add package Azure.Extensions.AspNetCore.Configuration.Secrets
dotnet add package Azure.Storage.Blobs
```

```
1g0y59RQhL3LMYad+ZgASm975PvJWft69q+jjFTo6Nyq5Sct0hPY1MVA0gN1qYIVN2oikVXDj==.
info : Installed Azure.Core 1.54.0 from https://api.nuget.org/v3/index.json to /Users/vaibhavgupta/.nuget/packages/azure.core/1.54.0 with content hash m6HbXlq9+GCBZ5A9kzFy1PdTwscX2APH7P1nqV4+yu+DH3RRtu
IDMRqdU17cMyinvHcPofpegoyQ6qWp7g==.
info : CACHE https://api.nuget.org/v3/vulnerabilities/index.json
info : CACHE https://api.nuget.org/v3-vulnerabilities/2026.07.01.00.04.29/vulnerability.base.json
info : CACHE https://api.nuget.org/v3-vulnerabilities/2026.07.01.00.04.29/2026.07.01.06.04.30/vulnerability.update.json
warn : NU1903: Package 'Microsoft.OpenApi' 2.0.0 has a known high severity vulnerability, https://github.com/advisories/GHSA-v5pm-xwqc-g5wc
info : Package 'Azure.Extensions.AspNetCore.Configuration.Secrets' is compatible with all the specified frameworks in project '/Users/vaibhavgupta/Desktop/Intern Work/genspark/Day 48 July 1/BlobVaultApi/BlobVaultApi.csproj'.
info : PackageReference for package 'Azure.Extensions.AspNetCore.Configuration.Secrets' version '1.5.1' added to file '/Users/vaibhavgupta/Desktop/Intern Work/genspark/Day 48 July 1/BlobVaultApi/BlobVaultApi.csproj'.
info : Generating MSBuild file /Users/vaibhavgupta/Desktop/Intern Work/genspark/Day 48 July 1/BlobVaultApi/obj/BlobVaultApi.csproj.nuget.g.targets.
info : Writing assets file to disk. Path: /Users/vaibhavgupta/Desktop/Intern Work/genspark/Day 48 July 1/BlobVaultApi/obj/project.assets.json
log : Restored /Users/vaibhavgupta/Desktop/Intern Work/genspark/Day 48 July 1/BlobVaultApi/BlobVaultApi.csproj (in 573 ms).
info : X.509 certificate chain validation will use the fallback certificate bundle at /usr/local/share/dotnet/sdk/10.0.203/trustedroots/codesignctl.pem'.
info : X.509 certificate chain validation will use the fallback certificate bundle at /usr/local/share/dotnet/sdk/10.0.203/trustedroots/timestampctl.pem'.
info : Adding PackageReference for package 'Azure.Storage.Blobs' into project '/Users/vaibhavgupta/Desktop/Intern Work/genspark/Day 48 July 1/BlobVaultApi/BlobVaultApi.csproj'.
info : GET https://api.nuget.org/v3/registration5-gz-semver2/azure.storage.blobs/index.json
info : OK https://api.nuget.org/v3/registration5-gz-semver2/azure.storage.blobs/index.json 324ms
info : Restoring packages for /Users/vaibhavgupta/Desktop/Intern Work/genspark/Day 48 July 1/BlobVaultApi/BlobVaultApi.csproj...
info : GET https://api.nuget.org/v3-flatcontainer/azure.storage.blobs/index.json
info : OK https://api.nuget.org/v3-flatcontainer/azure.storage.blobs/index.json 50ms
info : GET https://api.nuget.org/v3-flatcontainer/azure.storage.blobs/12.29.1/azure.storage.blobs.12.29.1.nupkg
info : OK https://api.nuget.org/v3-flatcontainer/azure.storage.blobs/12.29.1/azure.storage.blobs.12.29.1.nupkg 19ms
info : GET https://api.nuget.org/v3-flatcontainer/azure.storage.common/index.json
info : OK https://api.nuget.org/v3-flatcontainer/azure.storage.common/index.json
info : GET https://api.nuget.org/v3-flatcontainer/azure.core/1.55.0/azure.core.1.55.0.nupkg
info : OK https://api.nuget.org/v3-flatcontainer/azure.storage.common/index.json 24ms
info : GET https://api.nuget.org/v3-flatcontainer/azure.storage.common/12.28.0/azure.storage.common.12.28.0.nupkg
info : OK https://api.nuget.org/v3-flatcontainer/azure.core/1.55.0/azure.core.1.55.0.nupkg 54ms
info : OK https://api.nuget.org/v3-flatcontainer/azure.storage.common/12.28.0/azure.storage.common.12.28.0.nupkg 42ms
info : GET https://api.nuget.org/v3-flatcontainer/system.io.hashing/index.json
info : CACHE https://api.nuget.org/v3-flatcontainer/system.io.hashing/index.json
info : GET https://api.nuget.org/v3-flatcontainer/system.io.hashing/10.0.3/system.io.hashing.10.0.3.nupkg
info : OK https://api.nuget.org/v3-flatcontainer/system.io.hashing/10.0.3/system.io.hashing.10.0.3.nupkg 14ms
info : GET https://api.nuget.org/v3-flatcontainer/system.io.hashing/10.0.3/system.io.hashing.10.0.3.nupkg 41ms
info : OK https://api.nuget.org/v3-flatcontainer/system.io.hashing/10.0.3/system.io.hashing.10.0.3.nupkg 35ms
info : Installed System.IO.Hashing 10.0.3 from https://api.nuget.org/v3/index.json to /Users/vaibhavgupta/.nuget/packages/system.io.hashing/10.0.3 with content hash La6iCwsdTKhVX+LKN+pvFjQRR3LhLw3uKd12
knjLzRyPbSydF4cXidYx1iToDd6XVYdsBME18Zx1Z/OdIg==.
info : Installed Azure.Storage.Common 12.28.0 from https://api.nuget.org/v3/index.json to /Users/vaibhavgupta/.nuget/packages/azure.storage.common/12.28.0 with content hash 518YhNrks38zKLGFw2BBFLwIeyamH
/xauiBSAsdp2V7P98f+u2n22jKAmuQcALBupwH8nL2On8OyJ5Gw==.
info : Installed System.ClientModel 1.11.0 from https://api.nuget.org/v3/index.json to /Users/vaibhavgupta/.nuget/packages/system.clientmodel/1.11.0 with content hash 1W132zh7TbvN+HA08nQDuanA02Qu/0j0Nj
SrX0DtpUq5TX0Y33C6nd9/Ds21rgP4uMQYfXB35Q2fC21BhY==.
info : Installed Azure.Core 1.55.0 from https://api.nuget.org/v3/index.json to /Users/vaibhavgupta/.nuget/packages/azure.core/1.55.0 with content hash c7femvYS/xEuLp1nZN+DoQ2mx3X+G3ZX100z8RL/7cGMCM3JV
qepVhd3AW41K8AtT0S400igC0d4/zs9S0==.
info : Installed Azure.Storage.Blobs 12.29.1 from https://api.nuget.org/v3/index.json to /Users/vaibhavgupta/.nuget/packages/azure.storage.blobs/12.29.1 with content hash pmh7xZBt0Ycwi0853AB3uJ1TFVqs9P
Dte0Bu9/2z809mk1twe0Jj3vskMwUze01552f8v3PjetuQ0p8UQ==.
info : CACHE https://api.nuget.org/v3/vulnerabilities/index.json
info : CACHE https://api.nuget.org/v3-vulnerabilities/2026.07.01.00.04.29/vulnerability.base.json
info : CACHE https://api.nuget.org/v3-vulnerabilities/2026.07.01.00.04.29/2026.07.01.06.04.30/vulnerability.update.json
warn : NU1903: Package 'Microsoft.OpenApi' 2.0.0 has a known high severity vulnerability, https://github.com/advisories/GHSA-v5pm-xwqc-g5wc
info : Package 'Azure.Storage.Blobs' is compatible with all the specified frameworks in project '/Users/vaibhavgupta/Desktop/Intern Work/genspark/Day 48 July 1/BlobVaultApi/BlobVaultApi.csproj'.
info : PackageReference for package 'Azure.Storage.Blobs' version '12.29.1' added to file '/Users/vaibhavgupta/Desktop/Intern Work/genspark/Day 48 July 1/BlobVaultApi/BlobVaultApi.csproj'.
info : Generating MSBuild file /Users/vaibhavgupta/Desktop/Intern Work/genspark/Day 48 July 1/BlobVaultApi/obj/BlobVaultApi.csproj.nuget.g.targets.
info : Writing assets file to disk. Path: /Users/vaibhavgupta/Desktop/Intern Work/genspark/Day 48 July 1/BlobVaultApi/obj/project.assets.json
log : Restored /Users/vaibhavgupta/Desktop/Intern Work/genspark/Day 48 July 1/BlobVaultApi/BlobVaultApi.csproj (in 563 ms).
vaibhavgupta@F0998085F-vaibhavgupta BlobVaultApi %
```

#### Package

#### Purpose

|                                                          |                                                                                |
|----------------------------------------------------------|--------------------------------------------------------------------------------|
| <b>Azure.Identity</b>                                    | DefaultAzureCredential — uses your az login locally, Managed Identity in Azure |
| <b>Azure.Extensions.AspNetCore.Configuration.Secrets</b> | Maps Key Vault secrets directly into ASP.NET Core configuration                |
| <b>Azure.Storage.Blobs</b>                               | Blob Storage client for upload and download                                    |

## Lab 5: Wire Up Key Vault in Program.cs

---

### Step 1 — Edit Program.cs

8. Open BlobVaultApi/Program.cs in a text editor or IDE.
9. Replace the entire file content with the following:

```
using Azure.Identity;

var builder = WebApplication.CreateBuilder(args);

var keyVaultUri = new
Uri($"https://{builder.Configuration["KeyVaultName"]}.vault.azure.net/");
builder.Configuration.AddAzureKeyVault(keyVaultUri, new DefaultAzureCredential());

builder.Services.AddControllers();
builder.Services.AddEndpointsApiExplorer();
builder.Services.AddSwaggerGen();

var app = builder.Build();
app.UseSwagger();
app.UseSwaggerUI();
app.UseAuthorization();
app.MapControllers();
app.Run();
```

### Step 2 — Set the Vault Name in App Settings

10. Open BlobVaultApi/appsettings.Development.json and replace the KeyVaultName value with YOUR vault name (the value you wrote down from Lab 1):

```
{
  "KeyVaultName": "YOUR-KV-NAME-HERE",
  "Logging": { "LogLevel": { "Default": "Information" } }
}
```

#### Important

Replace YOUR-KV-NAME-HERE with the actual value from your KV\_NAME fill-in box on page 2. Using the wrong name will cause a connection error when you run the app.

### Step 3 — Verify the App Starts

```
dotnet run
```

Watch the startup log. You should see no errors. The app will print a localhost URL — leave it running.

```
vaibhavgupta@FVF0905WQ05F~vaibhavgupta BlobVaultApi % dotnet run
Using launch settings from /Users/vaibhavgupta/Desktop/Intern Work/genspark/Day 48 July 1/BlobVaultApi/Properties/launchSettings.json...
Building...
info: Microsoft.Hosting.Lifetime[14]
      Now listening on: http://localhost:5151
info: Microsoft.Hosting.Lifetime[0]
      Application started. Press Ctrl+C to shut down.
info: Microsoft.Hosting.Lifetime[0]
      Hosting environment: Development
info: Microsoft.Hosting.Lifetime[0]
      Content root path: /Users/vaibhavgupta/Desktop/Intern Work/genspark/Day 48 July 1/BlobVaultApi
```

## Lab 6: Blob Upload and Download Endpoints

---

### Step 1 — Register the Blob Service Client

11. Add the following to Program.cs, after the AddAzureKeyVault line and before builder.Build():

```
using Azure.Storage.Blobs;

builder.Services.AddSingleton(x =>
{
    var connStr = builder.Configuration["BlobStorageConnectionString"];
    return new BlobServiceClient(connStr);
});
```

### Step 2 — Create the Files Controller

12. Create a new file: BlobVaultApi/Controllers/FilesController.cs
13. Paste in the following code:

```
using Microsoft.AspNetCore.Mvc;
using Azure.Storage.Blobs;

namespace BlobVaultApi.Controllers;

[ApiController]
[Route("api/[controller]")]
public class FilesController : ControllerBase
{
    private readonly BlobContainerClient _container;
    private const string ContainerName = "uploads";

    public FilesController(BlobServiceClient blobServiceClient)
    {
        _container = blobServiceClient.GetBlobContainerClient(ContainerName);
        _container.CreateIfNotExists();
    }

    [HttpPost("upload")]
    public async Task<IActionResult> Upload(IFormFile file)
    {
        if (file is null || file.Length == 0)
            return BadRequest("No file provided.");

        var blobClient = _container.GetBlobClient(file.FileName);
        using var stream = file.OpenReadStream();
        await blobClient.UploadAsync(stream, overwrite: true);

        return Ok(new { fileName = file.FileName, url = blobClient.Uri.ToString() });
    }
}
```

```
[HttpGet("download/{fileName}")]
public async Task<IActionResult> Download(string fileName)
{
    var blobClient = _container.GetBlobClient(fileName);
    if (!await blobClient.ExistsAsync())
        return NotFound();

    var download = await blobClient.DownloadContentAsync();
    return File(download.Value.Content.ToArray(), "application/octet-stream",
fileName);
}
}
```

## Step 3 — Run and Test

14. Start the app:

```
dotnet run
```

15. Open a SECOND cmd.exe window (leave dotnet run running in the first). Test the upload endpoint — replace sample.jpg with any image or file you have:

```
curl -F "file=@sample.jpg" https://localhost:5001/api/files/upload -k
```

16. Test the download endpoint:

```
curl https://localhost:5001/api/files/download/sample.jpg -k -o downloaded.jpg
```

```
/dev/fd/12:25: command not found: compdef
vaibhav@vaibhav:~/Desktop/Intern Work/genspark/Day 48 July 1/BlobVaultApi$ dotnet run
vaibhav@vaibhav:~/Desktop/Intern Work/genspark/Day 48 July 1/BlobVaultApi$ curl -F "file=@Program.cs" http://localhost:5001/api/files/upload
{"fileName":"Program.cs","url":"https://c1l1b3a21280.blob.core.windows.net/uploads/Program.cs"}
vaibhav@vaibhav:~/Desktop/Intern Work/genspark/Day 48 July 1/BlobVaultApi$ curl -F "file=@sample.jpg" http://localhost:5001/api/files/upload
{"fileName":"sample.jpg","url":"https://c1l1b3a21280.blob.core.windows.net/uploads/sample.jpg"}
vaibhav@vaibhav:~/Desktop/Intern Work/genspark/Day 48 July 1/BlobVaultApi$ curl https://localhost:5001/api/files/download/sample.jpg -k -o downloaded.jpg
```

```
curl (35) LibreSSL/3.3.6: error:1404B42E:SSL routines:ST_CONNECT:tlsv1 alert protocol version
vaibhav@vaibhav:~/Desktop/Intern Work/genspark/Day 48 July 1/BlobVaultApi$ curl https://localhost:5001/api/files/download/sample.jpg -k -o downloaded.jpg
% Total % Received % Xferd Average Speed Time Time Time Current
Dload Upload Total Spent Left Speed
100 15996 100 15996 0 0 8451 0 0:00:01 0:00:01 --:-- 8450
vaibhav@vaibhav:~/Desktop/Intern Work/genspark/Day 48 July 1/BlobVaultApi$
```

## Lab 7: Deploy with a Managed Identity

### Step 1 — Create the App Service Plan and Web App

```
set APP_NAME=cli-lab-api-%SUFFIX%

az appservice plan create ^
  --name cli-lab-plan ^
  --resource-group %RG% ^
  --sku B1 ^
  --is-linux

az webapp create ^
  --name %APP_NAME% ^
  --resource-group %RG% ^
  --plan cli-lab-plan ^
  --runtime "DOTNETCORE:8.0"
```

#### Screenshot

*az webapp create output — JSON showing the web app hostname (e.g. cli-lab-api-XXXXX.azurewebsites.net)*

```
vaibhavgupta@FVF998WQ85F-vaibhavgupta BlobVaultApi % export LOCATION=southindia
vaibhavgupta@FVF998WQ85F-vaibhavgupta BlobVaultApi % az appservice plan create \
  --name cli-lab-plan \
  --resource-group $RG \
  --location $LOCATION \
  --sku B1 \
  --is-linux
[ Creating App Service Plan 'cli-lab-plan' (Linux).
App Service Plan create operation is throttled for subscription e25db867-f63d-489a-8bce-7de21e5f94a1. Please contact support if issue persists.
vaibhavgupta@FVF998WQ85F-vaibhavgupta BlobVaultApi %
vaibhavgupta@FVF998WQ85F-vaibhavgupta BlobVaultApi % ]
```

Prevented

APP\_NAME value

### Step 2 — Configure the App Setting

```
az webapp config appsettings set ^
  --name %APP_NAME% ^
  --resource-group %RG% ^
  --settings KeyVaultName=%KV_NAME%
```

### Step 3 — Assign a Managed Identity

```
az webapp identity assign --name %APP_NAME% --resource-group %RG%
```



### 17. Capture the identity's principal ID:

```
for /f "delims=" %a in ('az webapp identity show --name %APP_NAME% --resource-group %RG% -  
-query principalId -o tsv') do set PRINCIPAL_ID=%a
```



#### Screenshot

*az webapp identity assign output — JSON showing the principalId and tenantId*

## Step 4 — Grant the Identity Key Vault Access

```
az role assignment create ^  
  --role "Key Vault Secrets User" ^  
  --assignee %PRINCIPAL_ID% ^  
  --scope  
  "/subscriptions/%SUBSCRIPTION_ID%/resourceGroups/%RG%/providers/Microsoft.KeyVault/vaults/%KV_NAME%"
```

## Step 5 — Publish and Deploy

cmd.exe has no built-in zip command, so you will call PowerShell just for the compression step:

```
dotnet publish -c Release -o .\publish  
cd publish  
powershell -Command "Compress-Archive -Path * -DestinationPath ..\publish.zip -Force"  
cd ..  
  
az webapp deploy ^  
  --name %APP_NAME% ^  
  --resource-group %RG% ^  
  --src-path publish.zip ^  
  --type zip
```



#### Screenshot

*az webapp deploy output confirming successful zip deployment*

## Step 6 — Test in the Cloud

```
curl -F "file=@sample.jpg" https://%APP_NAME%.azurewebsites.net/api/files/upload
```



### Screenshot

*Successful cloud upload — curl output showing the blob URL on azurewebsites.net*

## Lab 8: Cleanup

---

### Important

This step permanently deletes all resources created today. Only run it when you are completely finished with the lab.

```
az group delete --name %RG% --yes --no-wait
```



### Screenshot

*az group delete running — terminal confirming the delete command was accepted*

## Reflection Questions

---

Answer the following questions to check your understanding of the lab.

1. Why is the Blob Storage connection string stored in Key Vault instead of in appsettings.json?

Key Vault keeps the connection string secure. It avoids storing sensitive information directly in the application files

2. What is DefaultAzureCredential, and why does the same line of code work both locally and in Azure?

DefaultAzureCredential automatically uses your Azure login on your computer and Managed Identity in Azure. So the same code works in both places.

3. Why does the App Service identity get the Key Vault Secrets User role instead of Secrets Officer?

The **Secrets User** role only allows reading secrets. It is safer because the app does not need permission to create or delete secrets.

4. What would happen if you accidentally committed appsettings.json with the raw connection string to a public GitHub repository? How does this lab's approach prevent that?

---

The **Secrets User** role only allows reading secrets. It is safer because the app does not need permission to create or delete secrets.

Common Errors

| Error                                                        | Fix                                                                                                                |
|--------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| <b>InvalidResourceGroup: name has invalid character '\$'</b> | You are using PowerShell/bash variable syntax. Switch to cmd.exe and use %VAR% syntax                              |
| <b>No such host known (vault.azure.net)</b>                  | The vault name in appsettings.Development.json is still the placeholder. Replace it with your actual KV_NAME value |
| <b>403 Forbidden (Key Vault)</b>                             | Role assignment hasn't propagated yet — wait 60 seconds and retry                                                  |
| <b>DefaultAzureCredential fails</b>                          | Run az login again in the same cmd.exe window                                                                      |
| <b>curl upload returns 401/403</b>                           | The App Service Managed Identity role assignment may not have propagated — wait 2 minutes                          |